

The image shows a MATLAB IDE window with three panes: Files, Editor, and Command Window.

Files Pane: Shows a folder named '5g and 6g' containing files 'harsha.m', 'untitled.m', and 'untitled2.m'. The Workspace pane below it lists variables: 'FrameTi...' (4.6150, 1x1), 'SlotDura...' (0.5769, 1x1), 'Slots' (8, 1x1), and 't' ([0,0.5769,1.1..., 1x9]).

Editor Pane: Contains the following MATLAB code:

```
1 clc; clear; close all;
2 FrameTime = 4.615; Slots = 8;
3 SlotDuration = FrameTime/Slots;
4 t = 0:SlotDuration:FrameTime;
5 figure
6 subplot(2,1,1)
7 stem(t,ones(size(t)),'filled')
8 title('TDMA Time Slot Structure')
9 xlabel('Time (ms)')
10 ylabel('Slot Level')
11 grid on
12 subplot(2,1,2)
13 plot(t,ones(size(t)),'-o','LineWidth',1.5)
14 title('Continuous View of Slot Allocation')
15
16 xlabel('Time (ms)')
17 ylabel('Slot Presence')
18 grid on
```

Figure 1: A 2x1 subplot figure titled 'Figure 1 X'.

- Top Plot: TDMA Time Slot Structure**
The y-axis is 'Slot Level' (0 to 1) and the x-axis is 'Time (ms)' (0 to 5). It shows a stem plot with blue filled circles at y=1 for time intervals [0, 0.5769], [0.5769, 1.1538], [1.1538, 1.7308], [1.7308, 2.3077], [2.3077, 2.8846], [2.8846, 3.4615], [3.4615, 4.0385], and [4.0385, 4.615].
- Bottom Plot: Continuous View of Slot Allocation**
The y-axis is 'Slot Presence' (0 to 2) and the x-axis is 'Time (ms)' (0 to 5). It shows a line plot with blue open circles at y=1 for the same time intervals as the top plot.

Command Window: Shows the prompt '>>'.

MATLAB

matlab.mathworks.com

Ask Gemini

HOME

PLOTS

APPS

FIGURE

FORMAT

TOOLS

Save As

Copy Figure

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Show Code

FILE

Title

Subtitle

X-Label

Y-Label

Legend

Colorbar

Grid

X-Grid

Y-Grid

Text Arrow

LABELS AND ANNOTATIONS

MATLAB Drive

untitled3 * x

untitled4 * x

5g and 6g

harsha.m

untitled.m

untitled2.m

Workspace

Name	Value	Size
Alloc	[1,1,1,1,1,0,0...	1x8
Blocked	[0,0,0,0,0,1,1...	1x8
TotalSlots	5	1x1
Users	[1,2,3,4,5,6,7...	1x8

8

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```
disp(table(Users',Alloc','VariableNames',{'User','Status(1=Allocated,0=Blocked)'))
figure
% Graph 1: Allocation per user
subplot(2,1,1)

stem(Users,Alloc,'filled','LineWidth',2)
title('TDMA Slot Allocation per User')
xlabel('User Index')
ylabel('Slot Status (1=Allocated, 0=Blocked)')
ylim([-0.2 1.2])
grid on

% Graph 2: System view (safe alternative to bar)
subplot(2,1,2)
plot(Users,Alloc,'-o','LineWidth',2)
hold on
plot(Users,Blocked,'-s','LineWidth',2)
title('TDMA Slot Utilization Analysis')
xlabel('User Index')
ylabel('Status')
legend('Allocated','Blocked')
grid on
```

Command Window

5	1
6	0
7	0
8	0

>>

Figure 1 x

TDMA Slot Allocation per User

User Index	Slot Status
1	1
2	1
3	1
4	1
5	1
6	0
7	0
8	0

TDMA Slot Utilization Analysis

User Index	Allocated	Blocked
1	1	0
2	1	0
3	1	0
4	1	0
5	1	0
6	0	1
7	0	1
8	0	1

Windows Taskbar

31°C Mostly cloudy

ENG IN

10:08:20 29-07-2026

```

1 clc; clear; close all;
2 TotalSlots = 8;
3 AllocatedSlots = 6;
4 FreeSlots = TotalSlots - AllocatedSlots;
5 Utilization = (AllocatedSlots/TotalSlots)*100;
6 disp('Frame Utilization (%)')
7 disp(Utilization)
8 figure
9 % Graph 1: Simple Line Plot (SAFE - no bar function)
10 subplot(2,1,1)
11 plot([1 2],[AllocatedSlots FreeSlots],'-o','LineWidth',2)
12 title('GSM Frame Utilization (Allocated vs Free)')
13 xlabel('Slot Type Index (1=Allocated, 2=Free)')
14 ylabel('Number of Slots')
15 grid on
16
17 % Graph 2: Pie Chart (SAFE)
18 subplot(2,1,2)
19 pie([AllocatedSlots FreeSlots])
20 title('Slot Distribution in GSM Frame')
21 legend('Allocated','Free')

```

